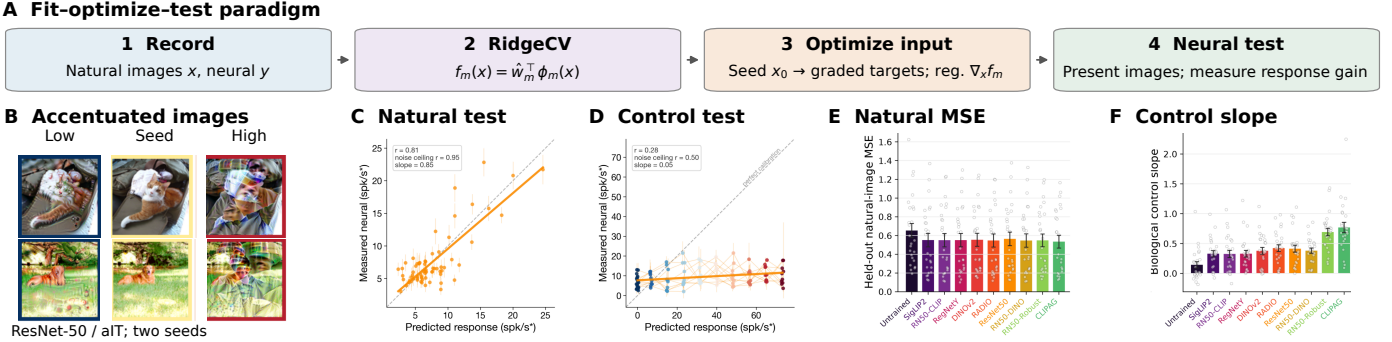

A Fit-optimize-test paradigm
1 Record
Natural images x, neural y
2 RidgeCV
$f_m(x) = \hat{w}_m^\top \phi_m(x)$
3 Optimize input
Seed x_0 → graded targets; reg. $\nabla_x f_m$
4 Neural test
Present images; measure response gain
B Accentuated images
Low
Seed
High
ResNet-50 / aIT; two seeds
C Natural test
r = 0.81
noise ceiling r = 0.95
slope = 0.85
Measured neural (spk/s*)
Predicted response (spk/s*)
D Control test
r = 0.28
noise ceiling r = 0.50
slope = 0.05
unified calibration
Measured neural (spk/s*)
Predicted response (spk/s*)
E Natural MSE
Held-out natural-image MSE
Untrained
SIgLIP2
RN50-CLIP
RegNetY
DINOv2
RADIO
ResNet50
RN50-DINO
RN50-RobustL
CLIPAG
F Control slope
Biological control slope
Untrained
SIgLIP2
RN50-CLIP
RegNetY
DINOv2
RADIO
ResNet50
RN50-DINO
RN50-RobustL
CLIPAG